

Validation of a Behind-the-meter BESS Control with degradation Cost by Hardware Experiments

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Abstract—To addresses the voltage issue in the distribution network caused by high PV penetration and the high cost of energy storage systems, a behind-the-meter battery scheduling and control algorithm has been proposed in this paper. The proposed solution includes a control framework that can schedule the day-ahead battery operation and prevent overvoltage in distribution systems. Two different techniques that aim to prolong battery life have been quantified, one is a hardware-based technique that uses a parallel connected supercapacitor to absorb large charging and discharging current; the other is a software-based technique that considers the battery degradation cost as part of the scheduling optimization. The proposed algorithm is evaluated via both the numerical simulation in the IEEE 13-node test feeder and hardware experiments. Based on the experiment results, a long-term system economic analysis has been made.

Index Terms—PV penetration, Battery energy storage system, optimal operation, Degradation

I. INTRODUCTION

The increasing renewable energies has developed rapidly in decades. Solar photovoltaic (PV) draws global attention due to its environmental and technical benefits. However, with the growth of the rooftop PV installation, several challenges have been found for the grid operator. One of the challenges caused by high PV penetration is due to the voltage regulation [1]. Although many researchers have proposed various potential solutions to address the voltage issues, battery energy storage system (BESS) seems to be a competitive candidate in the real word application. The installation of a proper sizing BESS compensating the fluctuation generated by PV system could benefit the onsite voltage. Because the BESS is owned by the customer and is located behind-the-meter, the possible economic benefits from the investment of BESS could be the energy charge saving and the demand charge reduction. Comparing with the energy charge, the demand charge has a higher portion in the monthly electricity bill in US. There are many studies emphasizing the advantages of demand charge reduction and providing scheduling algorithms for BESS [2][3].

In addition to the demand charge reduction, another potential manipulation to improve the BESS benefit is to prolong the

battery life. If the capacity reduces to 80% of the original value, then the battery needs to be replaced [4]. Extending the life of BESS will prolong the replacement period so that the investors receive higher profit. Although many researchers have been working on the battery degradation modelling [4][5][6], agreement has not been reached due to many reasons, parts of which could be the following: firstly, the battery materials are still in development, therefore, it is difficult to propose a mathematical equation that describes all different types of battery degradation behavior; secondly, the working condition of the BESS is related to the degradation rate as well; for example, the capacity of the battery cell will reduce quickly if the ambient temperature is high. Therefore, two techniques that potentially can prolong the battery life in the proposed application have been developed. One is called software-based technique, which considers the degradation cost in the scheduling algorithm, wherein we formulate the degradation cost as a general throughput degradation model. That general model can reflect the degradation cost to some degree but is inaccurate. Therefore, the practical battery life extension by is verified by hardware experiments. In addition to the consideration of the degradation mathematical model, an supercapacitor connected with the battery cell in parallel is also tested because that is a common structure used in electric vehicles (EVS) to increase the battery life by reducing the peak current [7]. The supercapacitor intergradation method is named hardware-based technique. To understand the improvement and effectiveness of both the mathematical degradation model and the parallel supercapacitor, the experiments test all combinations of them. With the degradation results, the replacement period of the BESS can be estimated. With the capital cost calculated, the long-term cost-benefit analysis of different configurations of the BESS has been estimated.

In this study, we propose a control algorithm that addresses the voltage issue caused by the PV penetration for BESS as the first task. The algorithm attempts the maximize the economic benefit for the customers as well. The cost of demand charge the degradation cost are introduced. Moreover, a hardware testbed is built to verify the degradation performance of the proposed algorithm in practical terms. Our contributions are listed as follows:

- We propose a scheduling algorithm that addresses the PV penetration issue for the utility and benefits the customers by considering behind-the-meter benefit such as demand charge reduction at the same time.

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- We verify the feasibility and evaluate the performance of the proposed algorithm by numerical simulation and hardware experiments.
- Two different techniques that can prolong the battery life in the behind-the-meter application has been tested. The long-term cost-benefits analysis is provided based on the experiment results.

The rest of the paper is organized as follows: Section II explains the objective function and constraints of the proposed algorithm. Section III provides numerical simulation results. In Section IV, the hardware experiment process and results are given. The long-term cost-benefits analysis is provided in this section. Finally, we draw our conclusion and some possible improvements through future work in Section V.

II. PROPOSED ALGORITHM

A. Objective function

The inputs of the proposed algorithm are the PV forecasting and load forecasting proposed in [8] and the output of the algorithm is the battery charge/discharge scheduling decision. Three different aspects are considered in the objective function: voltage variation, demand charge reduction and the degradation cost. The combination of these three terms builds the final objective function. The algorithm is similar to our previous work [2], the difference is that in this paper, there is no capacitor that provides reactive power to regulate voltage. Here we provide a brief introduction to the algorithm to make the paper comprehensible, more details can be found in our previous work.

Voltage variation is formulated as follows:

$$V_{\text{variation}} = \sum_{t=0}^T \|V_t - V_{\text{ref}}\|_2^2, \quad (1)$$

Where V_t is a vector that includes all load points' voltage magnitudes of at time t . V_{ref} means the reference voltage. $V_{\text{variation}}$ denotes voltage variation.

For the demand charge from the utility, the energy meter will record the power usage and energy usage for each 15-minute interval during a certain billing period for commercial and industrial customers. The highest power usage record is used to charge the customer as the demand charge. The total energy is also charged. However, the demand charge is usually much higher than the energy charge in the US. Therefore, only the demand charge is considered in this paper. Different utilities may have different ways to generate the bill. In this study, time-of-use (TOU) approach is applied [9]: during the off-peak hours (0:00 am to 10:00 am and 18:00 pm to 12:00 am), the demand charge is not counted; during the on-peak hours (10:00 am to 18:00 pm), the demand charge is counted. The demand charge is formulated as follows:

$$C_{\text{demand}} = r_{\text{demand}} \cdot \max(P_{\text{load},t} + P_{\text{PV},t} + P_{\text{bat},t}), \quad (2)$$

where C_{demand} is the total demand charge cost in the billing period and r_{demand} is the demand charge rate in \$/kW, $P_{\text{load},t}$ is the load power measured in the meter at time slot t . $P_{\text{PV},t}$ is the PV generation. $P_{\text{bat},t}$ is the battery charge/discharge power. The demand charge will be charged monthly, but (2) is

formulated daily because the daily PV forecasting and load forecasting are more accurate than the monthly forecasting. Although the daily optimum is no longer the monthly optimum, the problem is solved to some degree. In this study, we applied the time-of-use (TOU) framework for the demand charge, which means the demand charge only counts during the peak hours from 10:00 am to 18:00 pm.

A general throughput degradation cost is given in [4], and it can be regarded as a penalty cost for the operation the BESS. The controller discourages the operator to use the BESS in some sense.

$$C_d = \sum_{t=1}^T \frac{C_{\text{bat}}}{2 \cdot L_{\text{bat}}(\text{DOD}) \cdot \text{DOD}} \cdot |P_{\text{bat},t}| \cdot \Delta t, \quad (3)$$

where C_{bat} denotes the battery investment cost, $L_{\text{bat}}(\text{DOD})$ is an experiment-fitting function with specific depth of discharge (DOD) values, $P_{\text{bat},t}$ is the charging/ discharging power to the battery. Δt is 15-min interval

The total objective function is the weighted-sum of (1)-(3). In (2) and (3) the unit of the objective function is dollar, but the unit for (1) is volts or per-unit. Therefore, coefficients are needed to normalize different objectives.

$$\min J = \alpha V_{\text{variation}} + \beta C_{\text{demand}} + \gamma C_d, \quad (4)$$

where α , β and γ are coefficients. The unit of the demand charge and the degradation cost is the dollar. The voltage variation is in a per-unit system. The coefficients denote the weightings for different objectives. The settings of the coefficients depend on the operators' preference. If the operator would like to emphasize one of the objectives, the coefficients could be higher. During the off-peak hours, $\beta = 0$.

B. Constraints

For BESS, the active power charging/ discharging to the battery has a certain limit which is determined by the size of the inverter.

$$P_{\min} \leq P_{\text{bat},t} \leq P_{\max}, \quad (5)$$

where $P_{\max}$ and $P_{\min}$ denote the maximum and minimum power limits due to the physical constraints of the power electronics device.

The energy level of the BESS updates with the operating process,

$$E_{\text{bat},t} = \begin{cases} E_{\text{bat},t-1} + \eta_c P_{\text{bat},t}, & \text{if } P_{\text{bat},t} \geq 0 \\ E_{\text{bat},t-1} - \frac{P_{\text{bat},t}}{\eta_d}, & \text{if } P_{\text{bat},t} \leq 0 \end{cases}, \quad (6)$$

where η_c and η_d are the efficiencies of charging and discharging.

The energy level of BESS is also within a specific range.

$$E_{\min} \leq E_{\text{bat},t} \leq E_{\max}, \quad (7)$$

where $E_{\max}$ and $E_{\min}$ represent the upper and lower boundary of the energy levels of the battery.

For the distribution feeder, the voltage value should be managed in a safety range. In this study, the range is given as

$$0.95 \cdot I \leq V_t \leq 1.05 \cdot I, \quad (8)$$

where $\mathbf{1}$ is a vector consisting of ones. 0.95 and 1.05 are voltage magnitude boundaries in per unit value.

The overall optimization problem is formulated as (9).

$$\begin{aligned} \text{Min.} \quad & J = \alpha V_{\text{variation}} + \beta C_{\text{demand}} + \gamma C_d, \\ \text{subject to} \quad & P_{\min} \leq P_{\text{bat},t} \leq P_{\max}, \\ & E_{\text{bat},t} = \begin{cases} E_{\text{bat},t-1} + \eta_c P_{\text{bat},t}, & \text{if } P_{\text{bat},t} \geq 0 \\ E_{\text{bat},t-1} - \frac{P_{\text{bat},t}}{\eta_d}, & \text{if } P_{\text{bat},t} \leq 0 \end{cases} \\ & E_{\min} \leq E_{\text{bat},t} \leq E_{\max}, \\ & 0.95 \cdot \mathbf{1} \leq V_t \leq 1.05 \cdot \mathbf{1}, \end{aligned} \quad (9)$$

(1) is a 2-norm function, which is convex. (2) is the pointwise maximum function, which is known as convex. (3) is a linear function. Therefore, the objective function in (9) is a sum of convex functions, which is convex. As for the constraints, all equations in the constraints are linear. Therefore, the constraints are convex. We can conclude that the optimization problem is formulated as convex. The `cvx_toolbox` is applied to solve the optimization [10].

The controller requires the forecasting PV and load data as inputs to run the off-peak optimization and the on-peak optimization for different time slots. $P_{\text{bat},t}$ is the optimization variable.

III. NUMERICAL SIMULATION

In the numerical simulation, the proposed control algorithm is implemented in IEEE 13-node test feeder on node 675 phase B. The node 675B is assumed as an industrial customer with scaled PV panels. Both the load data and PV generation are shown in Figure 1.

The numerical cases are described as follows: For Case 1 and Case 2, only the voltage variation and the demand charge reduction take into consideration. In Case 3 and Case 4, the degradation cost is included. The only difference between Case 1 and Case 2 is that a supercapacitor is connected with the BESS in parallel and so do Case 3 and Case 4. As mentioned before, the supercapacitor is connected only with BESS and no control

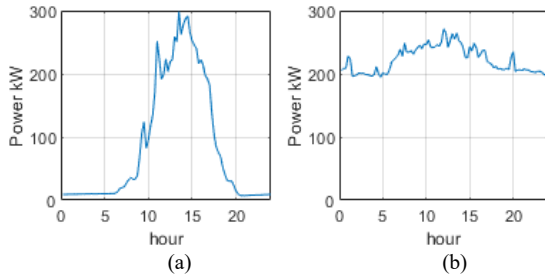


Figure 1. PV and load data: (a) Forecasting PV generation by 15-minute; (b) Forecasting load by 15-minute.

TABLE I.
SETTINGS FOR CASE STUDY

	Case 1	Case 2	Case 3	Case 4
α	1	1	1	1
β	1	1	1	1
γ	0	0	1	1
supercapacitor	No	Yes	No	Yes

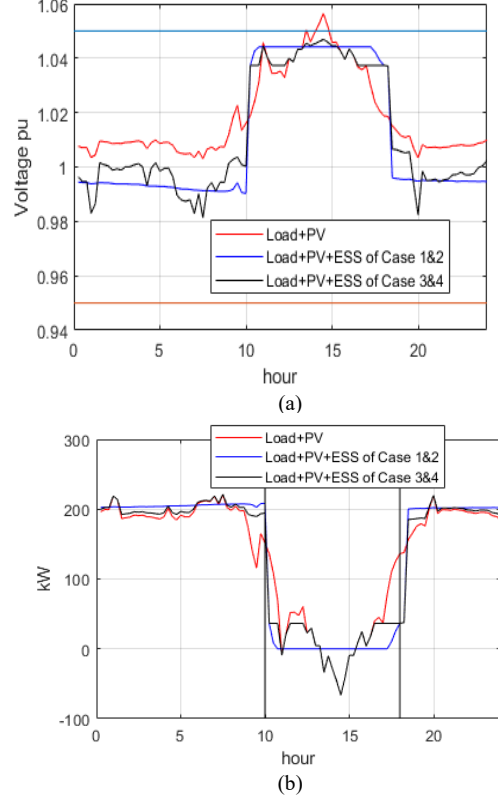


Figure 2. One-day performance results: (a) Voltage profile from Case 1 to Case 4; (b) Power usage from Case 1 and Case 4.

algorithm is applied. We assume that if the peak current comes, it will go automatically to the supercapacitor first and then the battery will follow the current smoothly. The existence of the supercapacitor will not affect the voltage and the power profile. TABLE I describes the setup for four cases. As stated before, different cases are setup to validate the improvement of the degradation model and the supercapacitor. There is no control action for the supercapacitor. Therefore, the numerical simulation cannot reflect the system level impact of the supercapacitor. In other words, Case 1 and Case 2 should be the same in this section, so do Case 3 and Case 4. Coefficient could be found in (9). If the coefficient is not zero, the corresponding objective is considered.

The BESS is designed as a 4-hour BESS with the 100 kW with 400 kWh. SOC limit is set as [0%, 100%]. The on-peak hours are 10:00 am to 18:00 pm and the rest are off-peak hours as mentioned before. Figure 2 gives the voltage profile results and power usage simulation for different cases. From the voltage profile in Figure 2(a), we can see the impact of the degradation cost on electric quality. The x-axis is the hours and y-axis denote the voltage in per unit. After considering the degradation cost in Case 3 and 4, the optimizer discourages the controller to use the batteries, so the voltage variation is larger during the on-peak hours. The voltage profiles in Case 1 and 2 have less variation than that in Case 3 and 4.

From the power usage results in Figure 2(b), the peak load value decreases from 100 kW in all cases. This reduced value is due to the BESS size. The x-axis is the hour, and the y-axis denote the power usage in kW. The power usage in the on-peak

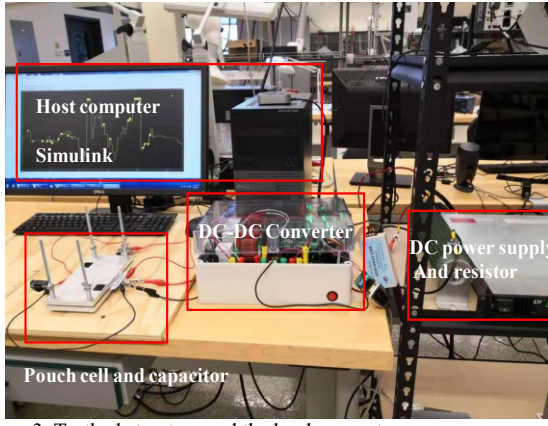


Figure 3. Testbed structure and the hardware setup.

period is less than the value in the off-peak period due to the PV generation during the on-peak hours. Note that these results are with the weightings in TABLE I. If the coefficients change, the power usage results vary. In the current setup, the results could ensure economic benefit to some degree. The numerical simulation can verify only the grid level feasibility such as voltage and power. As for the device degradation, the numerical simulation is insufficient. Therefore, it is necessary to build a hardware testbed to test the device performance of the battery degradation.

IV. BATTERY LIFE DEGRADATION STUDY

The high-power charge/discharge experiment requests high-power density on power electronic devices. We scaled the BESS size down to a pouch cell from a safety consideration. Because the materials of a small pouch cell and a large battery bank are the same, the degradation characteristic should be similar.

In this section, we will introduce our hardware experiments' process. The hardware testbed will be introduced. Then, the experiment process will be introduced. After that, the degradation test results will be shown. Based on the results, a 20-year system benefit estimation will be done.

A. Hardware testbed

The overall testbed can be seen in Figure 3. The testbed could generate customized current profiles. The operation power is controlled by the current and the terminal voltage values. The function of the charging/discharging function is achieved by the MATLAB/Simulink and Texas Instruments C2000 Processors, which requests several add-on toolboxes in MATLAB: Embedded Coder, Matlab Coder, and Simulink Coder. The DCDC converter is provided by Syndem.INC. With a Texas Instruments XDS100V2 USB JTAG Emulator to convey the setpoints from the MATLAB to the converter, the device can generate a 0-10A analog signal as the output. The setpoint of the charging/discharging process is grasped from the numerical optimization results. The DC power supply is AMETEK XG60-25. It has the following rating values: output voltage is from 0-60V; output current is from 0-25A; and a 5 Ω power resistor is used to provide the path to discharge. Because the BESS in the last section is a four-hour battery, which means the maximum charge/discharge C-rate is 0.25. The battery cell utilized in the hardware experiment is a pouch cell from Charge CCCV LLC.

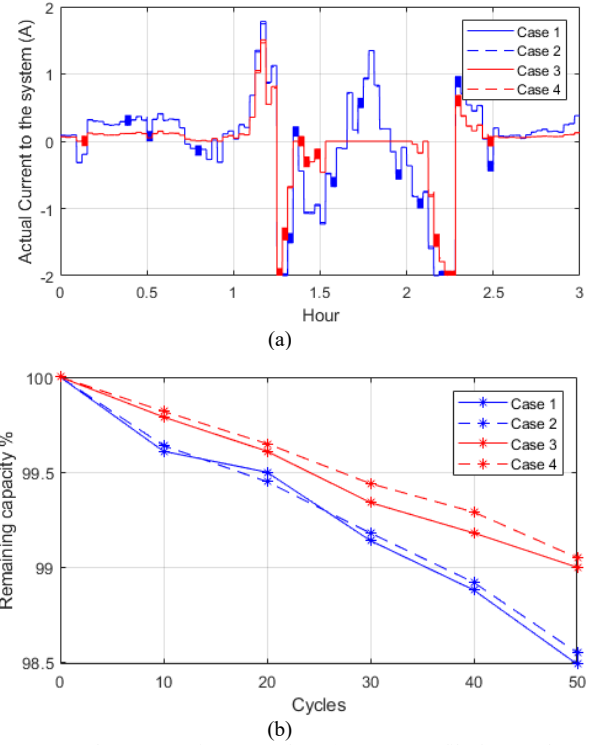


Figure 4. Hardware Experiment results: (a) current profile for pouch cells; (b) capacity fading of different cases.

The pouch cell is 1 Ah; therefore, the 1 C for this cell is 1 A. The maximum C-rate is 2 C and the voltage is from 2.8V to 4.2V. The supercapacitor is Taiyo Yuden with the voltage range from 2.2V to 4.4V. The capacitance is 100 F.

B. Experiment procedure and results

The hardware testbed can take the charge/discharge actions based on the optimal operating points given by the proposed algorithm. One problem for the experiment is that the degradation rate is not large which means if the 0.25 C-rate profile is used, it may take months to observe the capacity loss. Therefore, the accelerating battery life testing is used, which is introduced in our previous work [2]. We define the accelerated profile as one cycle, then the total experiment period is 50 cycles. After each 10-cycle, the remaining capacity of the pouch cell is measured. Figure 4 illustrates the capacity measurements results during the period. Case 1 has no software-based nor hardware-based techniques. Case 2 has the hardware-based technique but not the software-based method. Case 3 has both the softwarebased technique but not the hardware-based one. Case 4 has both the software-based and hardware-based approaches. It can be seen that the software-based technique is

TABLE II.
REMAINING CAPACITY OF THE POUCH CELL (ACCELERATED TEST)

	10 cycles	20 cycles	30 cycles	40 cycles	50 cycles
Case 1	99.61%	99.5%	99.14%	98.88%	98.50%
Case 2	99.64%	99.45%	99.18%	98.92%	98.55%
Case 3	99.79%	99.61%	99.34%	99.18%	99%
Case 4	99.8%	99.65%	99.44%	99.29%	99.02%

better than the hardware-based method in degradation reduction. TABLE II contains the information of the measured remaining capacity for each 10-cycle.

C. Long-term economic performance evaluation

To analyze the economic performance of the proposed algorithm in different cases, the first task is to map the accelerated test results to the practical battery capacity fading results. The major difference between the accelerated test and the practical operation is the C-rate. In [11], the authors reveal the relationship between the capacity loss and the number of cycles of both 2C and 1C condition. The capacity loss of 2C in 300 cycles is the same as the capacity loss of 1C in 500 cycles. In [12], the capacity loss of 0.25C in 100 cycles is the same as the capacity loss of 1C in 50 cycles. After combining the results, the 10 cycles of 0.25C operation is about equal to the 3 cycles of 2C operation in terms of capacity loss. Based on that, the replacement period could be estimated. For example, the capacity loss in our 50-cycle accelerated test is 1%. If the capacity loss rate remains the same, it takes 1000 cycles when the cell capacity reaches 80%. After converting to 0.25C operation, the equivalent cycles are $1000 \times 10/3 = 3333$. One equivalent cycle means one-day operation. Therefore, 2,200 cycles (days) equal 9.1 years.

Based on the report [13] from the National Renewable Energy Laboratory, the capital cost of the battery energy storage system is 209 \$/kWh. In a PNNL annual report of 2019 [14], the capital cost of the supercapacitor is 160 \$/kW. The sizing of the BESS in this report is 100 kW and 400 kWh. The sizing of the supercapacitor is 100 kW as well. The total capital cost of the battery system is $209 \times 400 = \$ 83,600$; the capital cost of the supercapacitor system is $160 \times 100 = \$ 16,000$.

Assume that the used daily load data is a typical load pattern and as does the PV. Define the system benefit as the total demand charge savings minus the sum of capital cost and replacement cost. The demand charge saving can be found in numerical simulation results. The total system benefit for 20-year period can be given in TABLE III. Because the total demand charge saving is related to the local demand charge rate, we select several different cities in New York State to evaluate the system economic performance. From the TABLE III, it is clear that when the demand charge rate is high, the system results in a satisfying economic performance. In addition to, it is clear that Case 3 always obtain the highest benefit among all cases. The reason is the improvement for prolonging the

replacement period by adding the supercapacitor is not enough. On the other hand, the supercapacitor also requires the capital cost; therefore, the economic benefit will be reduced. The potential reason for the insufficient improvement might be that, the extreme peak current may not happen as frequently as EV conditions or the capacitance is inadequate.

V. CONCLUSION

In this paper, we proposed a scheduling algorithm for BESS. The proposed method addresses the voltage issue caused by PV penetration and the objective function contains the demand charge reduction and degradation cost as well. The algorithm is verified in both numerical simulation and hardware experiment. Based on the results, the long-term economic performance is evaluated. Future study includes the compensation of the forecasting mismatch, accurately mapping between the accelerate battery life test and the practical low C-rate condition and proving the economic evaluation by using more data.

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TABLE III.
20-YEAR SYSTEM INCOME ESTIMATION (\$) FOR DIFFERENT CITIES IN
NEW YORK STATE

	Long Island	NYC	Rochester	Albany	Buffalo
Demand charge rate	49.43 \$/kW	24.51 \$/kW	16.24 \$/kW	11.92 \$/kW	10.62 \$/kW
Case 1	851,920	253,840	55,360	-48,320	-79,520
Case 2	835,920	237,840	39,360	-64,320	-95,520
Case 3	935,520	337,440	138,960	35,280	4,080
Case 4	919,520	321,440	122,960	19,280	-11,920